

# LOCUS – AI-Powered Wearable Memory & Cognitive Assistant

## 1. Intelligent Routine Learning & Anomaly Detection Pattern Recognition System

- **Daily Pattern Learning:** Automatically learns user's regular routines and habits.
  - "You usually take your keys when leaving home at 8 AM."
  - "You typically take medication with breakfast at 7:30 AM."
  - "Regular evening walk between 6–7 PM."

### Proactive Alert System

- **Behavioral Anomaly Detection:** Identifies deviations from normal patterns.
  - "You're leaving without your wallet today, intentional?"
  - "Unusual inactivity detected this morning."
  - "Missed your medication time."

### Predictive Assistance

- **Context-Aware Suggestions:** Provides reminders based on learned patterns.
  - "Don't forget an umbrella—rain expected this afternoon."
  - "Time to leave for your weekly grocery shopping."

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## 2. Multi-Modal Memory Recall System Advanced Search Capabilities

- **Visual Search:** "Show me where I saw my passport last week."
- **Audio Context Search:** "What was I talking about when I had my keys yesterday?"
- **Temporal Filtering:** "Where did I put my glasses this morning?"
- **Relationship Mapping:** Connects objects, people, locations, and times.

### Cross-Reference Memory Access

- **Unified Memory Database:** All memories interconnected and searchable.
- **Natural Language Queries:** Conversational search.
- **Timeline Visualization:** Chronological memory review.

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## 3. Social Interaction Assistant Face Recognition & Context

- **Smart Identification:** "This is Dr. Smith, your cardiologist from last month."
- **Relationship Database:** Stores context about people.
- **Interaction History:** "Last time you met Sarah, you discussed the family reunion."

### Discreet Social Support

- **Real-Time Prompts:** Subtle name reminders.
- **Conversation Starters:** Topic suggestions.

- **Contextual Reminders:** "John's birthday is next week."
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## 4. Caregiver Support System

### Family Dashboard

- **Accessible Web Interface** for authorized caregivers.
- **Real-time Activity Monitoring**
- **Medication Tracking**

### Remote Assistance

- Send Reminders Remotely
  - Status Updates
  - Emergency Alerts
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## 5. Cognitive Training & Memory Games Suite

### Personalized Brain Training

- Spatial memory grid
- Custom quizzes based on frequently forgotten items
- Pattern recognition
- Word association exercises

### Adaptive Learning

- Number sequence challenges
  - Face-name matching
  - Automatic difficulty adjustment
  - Progress tracking
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## 6. Smart Medication Adherence System

### Multi-Method Verification

- Visual confirmation
- Automated tracking
- Dosage verification

### Intelligent Reminders

- Escalating alert protocol (gentle → urgent → notify family)
- Context-aware timing
- Flexible snooze options

### Caregiver Portal

- Real-time medication status
- Missed-dose alerts
- Weekly compliance reports
- Remote override support

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## 7. Advanced Location Safety System

### Real-Time Location Monitoring

- Live position sharing
- Safe zone management
- Geofencing alerts
- Movement history

### Emergency "I'm Lost" Mode

- Panic button
- Automatic wandering detection
- Voice-activated help
- Continuous location sharing
- Two-way communication

### Guided Navigation Home

- Step-by-step directions
- Landmark-based navigation
- Accessible interface
- Offline support

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## 8. Privacy & Security Center

### Privacy Protection

- Automatic face blurring
- Configurable data expiry
- Recording pause button
- Selective memory storage

### Security Features

- End-to-end encryption
- PIN/Fingerprint access
- Multiple privacy modes

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## 9. ADHD Support

- Time blindness
- Task switching difficulty
- Losing items frequently

### Added Features

- Smart Task Chunking: Break tasks into smaller actionable steps.
- Focus Timer (Pomodoro Mode): Helps maintain focus.
- Hyperfocus Alerts: Detects when user is stuck on a single activity for too long.
- Distractibility Monitoring: Movement + phone usage pattern analysis.

- **Reward & Gamification System:** Encourages task completion.
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## 10. Depression Support

- Low Motivation
- Inactivity
- Disrupted Routines
- Withdrawal Symptoms

### Added Features

- **Mood Check-in System:** Simple daily emotional tracking.
  - **Inactivity Alerts:** "You haven't moved much in the last hour. Want to stretch?"
  - **Positive Prompting System:** Encouragement micro-reminders.
  - **Early Warning Behavioral Flags:** Notifies family/caregivers of concerning patterns.
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## 11. Autism/ASD & Special Cognitive Conditions Support

- Social cue difficulty
- Routine dependency
- Sensory overload

### Added Features

- **Conversation Cue Suggestions:** Prompts for what to say next.
  - **Noise/Sensory Overload Detection:** Alerts if environment becomes overwhelming.
  - **Strict Routine Protection:** Alerts when routine deviates.
  - **Safe Spaces Map:** User-defined quiet/safe locations.
  - **Behavior Shift Detection for Bipolar Disorder:** Sudden routine changes flagged.
  - **Sleep Cycle Monitoring:** Detects irregularities.
  - **Energy Pattern Analysis:** Movement and speech-based indicators.
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## 12. Normal Users Forgetfulness Support

- Misplacing everyday items
- Forgetting appointments or tasks
- Inconsistent daily routines

### Added Features

- **Smart Reminder System:** For keys, wallet, phone, etc.
  - **Visual/Auditory Prompts:** Gentle alerts for missed tasks.
  - **Habit Formation Tracking:** Encourages routine consistency.
  - **Quick Search Feature:** Helps locate commonly misplaced items.
  - **Daily Checklist:** Allows user to mark completed tasks.
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## **13. Technical Architecture Extensions Condition-Specific Behavior Modeling**

- Separate behavioral models for ADHD, depression, ASD, bipolar disorder, dementia.
- Uses routine patterns, movement profiles, sleep analysis, social interaction frequency.

### **Recommendation Engine**

- Delivers different advice for different conditions.
- Embeddings + rule-based hybrid system.

### **Unified Caregiver Dashboard**

- Condition-specific alerts
  - Personalized insights
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